

MEMO — Milestone 1: EDA & Volatility Regime Audit

Swiss Market Forecasting Project | UBSG, NESN, NOVN, ROP (Roche) vs. SMI | Data vintage: 2026-07-24 | Revision v2 (post-review)

RECOMMENDATION

Proceed to Milestone 2 (GJR-GARCH(1,1), Student-t innovations) as architected. Data is clean and audited. Two conditions carried forward: (1) treat UBSG with separate volatility/regime handling given its distinct rate sensitivity and a real, though sample-limited, FX-beta shift around the Credit Suisse merger; (2) do not winsorize the NESN 2026-07-23 earnings-day observation — keep it, the GJR-GARCH/Student-t architecture exists precisely to capture events like this one.

KEY FINDINGS

Item	Finding	Evidence
Stationarity	All 5 return series (UBSG, NESN, NOVN, ROP, SMI) confirmed stationary — ADF and KPSS agree.	ADF $p < 0.001$ all names; KPSS $p > 0.10$ all names
Correlation — full sample	Volatility correlation (0.55–0.85) runs consistently higher than return correlation (0.30–0.60).	See Section 6
Correlation — by regime (corrected, r2)	2008/2015/2020 show the expected stress-convergence pattern. 2022 does NOT — correlations were lower than the calm baseline.	Section 7 — regime-conditioned matrices
UBSG FX-beta, CS merger split (corrected)	USDCHF beta roughly halves ($-0.14 \rightarrow -0.07$) and loses significance post-merger (12-Jun-2023); EURCHF beta stable. Real but sample-limited.	Appendix A.1
NESN earnings-day event (corrected)	-7.98% (~ -7.2 std devs), confirmed genuine vs. external sources. Recommendation: annotate-and-keep. Elevates 21d vol through ~ 20 -Aug.	Section 3
Sandoz spin-off (NOVN)	Value transfer NOT encoded in price series (adj/unadj ratio constant at 0.903011 through event). Annotated, not corrected.	Section 5
Roche ticker conversion (ROG $\rightarrow$ ROP)	Legal reclassification only, no value-transfer gap. Continuous history confirmed against Roche's own published year-end prices.	Data audit, prior review
HTML export bug (corrected)	Notebook images now embedded as base64 cell outputs, not markdown links to relative paths — survives export outside the repo folder.	Verified: 0 broken links in exported HTML

LIMITATIONS

Common-sample correlation/stationarity work constrained to 2003–present by FX data availability. Regime windows are manually specified, not statistically detected. UBSG post-merger FX-beta split rests on ~ 2 years of data (778 obs) — treat as provisional pending more post-merger history. FX-beta regressions are simple contemporaneous OLS, no lag structure — adequate for EDA, not for Milestone 2.

NEXT STEP

Milestone 2: GJR-GARCH(1,1), Student-t innovations. 1-day VaR via direct conditional-variance forecast; 10-day stress via Monte Carlo forward simulation (not sqrt-h scaling), filtered historical simulation as cross-check. UBSG evaluated for separate-regime calibration. ML challenger model deferred to Milestone 3. Backlog (not in Milestone 2 core): test what actually drives the 2022 decorrelation finding — rolling intra-2022 correlation vs. rate-decision dates, to distinguish differentiated-repricing from idiosyncratic/noise explanations.

Full detail, code, and regime-conditioned/split-sample tables: `notebooks/01_eda_regime_correlation.ipynb`. Figures: `reports/figures/`.